

Brain Document Ingestion

How the platform sends an uploaded document into Brain and reads back what Brain understood from it. The ingestion architecture is real and tested. This spec is the exact contract to build against, plus the preconditions that must land before the document path works in a hosted environment.

STAGING `https://api.sandbox.brain.fi/v1`

PROD `https://api.brain.fi/v1`

- **Status: build against this now, but do not expect live results yet**

The storage, normalization, ledger, policy gate, and question layers are built, tested, and deployed to staging inside the `api` service. The document extraction step that turns an uploaded file into structured data lives in a separate service that is not yet built or deployed by CI. Three brain-core preconditions (Section 3) must land before an upload produces a readable obligation. The API contract itself is stable and safe to code against today.

01 THE PIPELINE, END TO END

Six steps. The platform initiates three of them over the public API. The rest run automatically inside Brain. Each step is marked with its current deployment state.

1

Upload the document ● DEPLOYED

platform to brain POST `/v1/raw/ingest` as multipart form data with the file, its mime type, and a `source_type` of `pdf_upload` or `csv_upload`. Brain stores the raw bytes immutably and returns a `raw_id`. Content addressed, so re-uploading the same file de-duplicates.

2

Extract text and fields ● ENDPOINT TO BE BUILT

platform to brain POST /v1/raw/{raw_id}/extract . This public trigger does not exist yet. Brain reads the stored bytes, runs deterministic text extraction (CSV, XLSX, digital PDF) then a model pass to pull the counterparty, amount, due date, and direction, and writes a parsed record. Brain does this internally by calling its own extraction service. The platform never calls that service directly.

3

Promote to an obligation

• AUTOMATIC

• DEPLOYED

inside brain A worker polls every 15 seconds, projects the parsed record through the canonical layer, and materializes a ledger obligation. Model-read data is written low trust: provenance agent_contributed , confidence capped at 0.5 . No platform action required.

4

The gate refuses to pay on it alone

• BY DESIGN

• DEPLOYED

inside brain A document read on its own can never authorize a payment. The payment gate reads the ledger only and rejects auto-execution on low-trust document evidence until an independent source corroborates it or a human confirms. This is the safety property, not a limitation. See Section 6.

5

Read back what was found

• DEPLOYED

platform to brain GET /v1/ledger/obligations returns the structured payables and receivables Brain extracted. This replaces the mock results screen with real data: who is owed, how much, when, and at what confidence.

6

Ask questions about it

• DEPLOYED

platform to brain POST /v1/wiki/question answers plain-language questions grounded in those obligations: what is due, what is overdue, what is owed to a given vendor. Returns an answer plus the evidence rows it used.

02 AUTH AND THE ONE HARD RULE

The platform talks only to the public api service under /v1 . It authenticates with a tenant-scoped token carrying the scopes below. Every other Brain service is internal only and has no public ingress.

SCOPES THE PLATFORM TENANT NEEDS

- `raw:write` to upload and to trigger extraction
- `ledger:read` to list obligations and accounts
- `wiki:read` to ask grounded questions

THE RULE THAT KEEPS THIS SAFE

- Never call the extraction service directly. It is internal, separately authenticated, and not exposed. Extraction is orchestrated by Brain behind the public `/v1/raw/{id}/extract` trigger.
- Treat extracted data as advisory. It is capped at **0.5 confidence** and cannot move money on its own.

03 PRECONDITIONS BEFORE THE DOCUMENT PATH WORKS LIVE

These are brain-core tasks, not platform tasks. Until they land, an upload on staging stores the file but does not produce an obligation. The platform can build and test its calls against the contract in parallel.

Deploy the extraction service **BRAIN-CORE**

The document extractor exists and its unit tests pass, but CI does not build or deploy it. Add it to the image build matrix and the staging and production deploy steps, with its model API key and the shared signing secret provisioned.

Add the public extraction trigger **BRAIN-CORE**

Step 2 above. A public `POST /v1/raw/{id}/extract` route on the api service that loads the stored bytes and calls the extraction service internally over a signed request. Nothing triggers extraction today.

Promote to production **BRAIN-CORE**

Production runs a behind revision. A merge to main reaches staging automatically but production requires a manual promote. Point early integration at staging, and confirm each route returns non-404 on the target host before wiring it.

One known limitation beyond these: scanned images and photos of documents need an OCR step that is deferred. Digital PDFs, CSV, and XLSX exports work today. A photographed insurance card or a scanned statement will not extract until OCR lands.

04 ENDPOINT CONTRACTS

POST /v1/raw/ingest

scope **raw:write** / ● **DEPLOYED**

BODY multipart form data: `source_type` (`pdf_upload` or `csv_upload`), `file` , optional `mime_type` . Max 50 MB.

RETURNS `{ raw_id, sha256, deduplicated }` . Status 201 for a new artifact, 200 if the identical file already existed.

POST /v1/raw/{raw_id}/extract

scope **raw:write** / ● **TO BE BUILT**

BODY `{}` . The raw id in the path is sufficient. Brain resolves the mime type from the stored artifact.

RETURNS `{ parsed_id, confidence }` . Idempotent: re-posting for the same artifact returns the existing parsed record.

GET /v1/ledger/obligations

scope **ledger:read** / ● **DEPLOYED**

QUERY optional `status` (`upcoming` , `due` , `overdue`), `type` , `due_before` .

RETURNS `{ obligations: [ { id, type, direction, amount_due, currency, due_date, status, counterparty_id, provenance, confidence } ] }`

POST /v1/wiki/question

scope **wiki:read** / ● **DEPLOYED**

BODY `{ question: "what do I owe and when is it due" }`

RETURNS `{ answer, evidence: [...] }` . The answer is grounded in the ledger obligations above, with the evidence rows it drew on.

05 WHO BUILDS WHAT

PLATFORM BUILDS

- The four public API calls above, in sequence.
- Replace the mock reading and results screens with real state: poll on `raw_id`, then render **ledger obligations**.
- Show **confidence** on extracted items and label them advisory. Never present a document read as ready to pay.
- An email forwarding path is available too: `source_type` `email_inbound` lets a user forward a document instead of dragging it in.

BRAIN-CORE SHIPS FIRST

- Build and deploy the **extraction service** in CI.
- Add the public `/v1/raw/{id}/extract` trigger.
- **Promote to production** once verified on staging.
- Later: **OCR** for scanned and photographed documents.

06 THE TRUST MODEL, SO THE PLATFORM FRAMES IT RIGHT

A document informs what Brain proposes. It never authorizes what Brain executes.

Everything a model reads out of an uploaded file is written at low trust and capped at 0.5 confidence. The payment gate reads the ledger only. It cannot read this data as authoritative, so a document read on its own can never move money. Data earns the right to drive an action through corroboration from an independent source, or through an explicit human confirm.

document read to proposal, corroboration or confirm to authorization

For the platform this means: surface extracted obligations as insight and awareness (what is due, what is overdue, anomalies), and route any payment action through Brain's normal approval flow. Do not build a separate path from document to payment. That path would bypass the one guarantee the architecture exists to hold.

07 FIRST CALL TO MAKE

Once staging has the extraction service and the trigger, the whole loop is four requests:

```
# 1. upload
POST /v1/raw/ingest      multipart: source_type=pdf_upload, file, mime_type
  → { raw_id }

# 2. extract
POST /v1/raw/{raw_id}/extract {}
  → { parsed_id, confidence }

# 3. read (obligation appears within ~15s)
GET /v1/ledger/obligations?status=due
  → { obligations: [ ... ] }

# 4. ask
POST /v1/wiki/question   { "question": "what is overdue" }
  → { answer, evidence }
```

Brain Document Ingestion / Integration Spec

Verified against brain-core main. Confirm on staging before wiring production.